Technical Test Task: Architectural Image Relighting & Sun Direction

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Summary

Four tools, three photos (old barn, sunset building, interior), free options only, 1–3 runs per tool and photo: Sun Direction LoRA and IC-Light (Hugging Face Spaces), Qwen Relight LoRA (ComfyUI Cloud) and the commercial tool Relight Spaces..

- **Key finding.** No tool both moved the sun and kept the scene intact. The one clear move of the light was Sun Direction LoRA on the interior (6° elevation); every other result changed color, sky or brightness while shadows stayed in place.
- **Sun Direction LoRA (13–17 s).** Best direction control, but it adds objects (second chimney), switches lamps on, greys colors, can give a blank white sky and returns 57–71% of the input width. Base model is non-commercial.
- **Qwen Relight LoRA (5–44 s).** Kept geometry and lamps best of the open tools on the barn and interior and is open-source with commercial use support, but did not move the sun there. On the sunset building a mild prompt kept the golden hour, a strong midday prompt invented a second house, a strict prompt changed almost nothing.
- **IC-Light (5–7 s).** Fastest, but crops to 4:5 (512×640), did not change the sun, greyed the barn wall and only repainted the building's sky. Not usable for whole buildings as tested.
- **Relight Spaces (30–39 s, from $0.40 per photo).** Fixed presets, no sun-direction control. Best time-of-day change and preservation: the sunset building became a clear blue-sky day with every object in place at full size; on the barn and interior it only warmed the light.
- **Reliability.** Free demos were unreliable: Kontext Relight LoRA could not be tested (Space paused, copies errored) and the Sun Direction Space ran out of GPU quota.

Results at a glance:

Tool	Sun direction moved?	Scene kept?	Time	Output size	Cost / license
Sun Direction LoRA	Interior yes (6°); weak on barn, building	Mostly; extra chimney, lamps on	13–17 s	57–71% of input width	Free; base non-commercial
Qwen Relight LoRA	No (color and contrast only)	Barn, interior yes; building depends on prompt	5–44 s	57–73% of input width	Free tier; open-source with commercial use support
IC-Light	No	Building yes; barn wall greyed	5–7 s	512×640 (4:5 crop)	Free Space; repo open-source with commercial use support
Relight Spaces	No (time of day via presets)	Yes, all three photos	30–39 s	Same size or larger	$0.40–0.75 per photo

Recommendation: Do not ship any tested option as a “sun direction” feature yet. For time-of-day presets, build on Qwen-Image-Edit-2509 + Relight LoRA on our own GPU (open-source model), use Relight Spaces as the quality bar and hosted fallback, and gate every output with automatic checks. Treat real sun-direction control as R&D: train a LoRA on an open-source model using renders with known sun positions.

1. Market Overview & Tool Selection

I looked at four categories of tools that can change light direction or time of day in an image: open models, LoRAs, ComfyUI workflows, and APIs / commercial tools. The table lists what is worth trying, with one line on each. Facts were checked on the linked pages in October 2026; items I could not confirm are marked.

Tool	What it does (one line)	License / cost
IC-Light Open model	SD1.5-based relighting model (“Imposing Consistent Light”) conditioned on text and a foreground image; direction is set via an initial-latent “Lighting Preference” (left/right/top/bottom). Demos focus on portraits and objects, so whole buildings are a stress test. GitHub | HF Space	Repo: open-source with commercial use support. README warns the bundled BRIA RMBG 1.4 background remover is non-commercial; replace with BiRefNet for commercial use.
Sun Direction LoRA (FLUX.2 klein 9B) LoRA	Image-to-image LoRA for controlling sun direction and elevation on exterior images, the closest match to this task. Input: Space: sun-position control (e.g. top-left, bottom-left) plus optional extra prompt, as used in my tests. Model | Demo Space	LoRA: open-source with commercial use support. Base FLUX.2 klein 9B: non-commercial base model, so not usable commercially as-is. (klein 4B is open-source with commercial use support.)
Relighting Kontext Dev LoRA v3 LoRA	Prompt-driven relighting on FLUX.1 Kontext [dev]. The demo Space has presets (e.g. “golden time”, “sunset over sea”) and direction hints such as “top left”. Not tested: the original Space was paused and the copies I tried returned errors. Demo Space	LoRA license is listed as “other” on its model page; the base FLUX.1 Kontext [dev] is non-commercial.
Qwen-Image-Edit-2509 Relight (dx8152) LoRA	Relight LoRA for Qwen-Image-Edit-2509, driven by a text prompt. Available in the multi-LoRA Qwen-Image-Diffusion Space. The same author also has a “Light Migration” LoRA that copies lighting from a reference image.	LoRA and base model Qwen-Image-Edit-2509: both open-source with commercial use support (model cards, checked Oct 2026). The model card asks to combine it with a Lightning LoRA (license not checked).
ComfyUI-IC-Light (Kijai nodes) ComfyUI workflow	IC-Light inside ComfyUI: mask-based control of where the light comes from, optional IP-Adapter for style. Finest control, but needs a local or cloud GPU.	Same as IC-Light.
IC-Light V2 on fal.ai API	Hosted IC-Light V2 for relighting and background change; listed for commercial use. fal.ai	$0.10 per megapixel (checked Oct 2026).
Magnific Image Relight API Commercial API	Hosted relight API from Magnific. Docs	€0.10 per operation (checked Oct 2026).

Tool	What it does (one line)	License / cost
Relight Spaces Commercial web tool	Web app for property photos with 14 fixed lighting presets (twilight, sunny, warm interiors). No sun-direction control. Returns images at the photo's own size and re-checks house numbers and signs after relighting. No API mentioned on the site (not verified). relightspaces.com	$0.40–0.75 per photo (one-time credit packs); 3 free relights (checked Oct 2026).

2. Test Images

Three images from Unsplash, chosen to cover different failure modes:

- **Image 1, old barn exterior:** thatched roof texture, wooden details and hard shadows from low autumn sun. Unsplash photo, photographer @onegrandtrip.
- **Image 2, modern timber building at sunset:** near-silhouette with strong orange side light; tests a drastic time-of-day change. Unsplash photo, photographer @itsandreaswav.
- **Image 3, interior with window blinds:** thin repeating blind lines and light patterns on the floor; tests interior shadow casting and fine geometry. This is outside the intended domain of the Sun Direction LoRA (exteriors), so a failure is expected and informative. Unsplash photo, photographer @isaax_the_artist.

3. Evaluation Methodology & Metrics

Every result is scored 1–5 on four criteria (Geom. = geometry, Mat. = material, Light = light change, Shad. = shadow realism), plus time per image in seconds:

- **Geometry preservation:** walls, windows, roof lines, blind slats stay in the same positions. Objective check: share of Canny edges of the original that also appear in the result (5 px tolerance).
- **Material consistency:** wood, thatch, glass and plaster keep their texture and base color.
- **Light change:** the light visibly comes from the requested direction / time of day.
- **Shadow realism:** shadows are consistent with the new light direction and with each other.
- **Time and cost per image:** time as shown on the Space or measured with a timer; cost from the provider's price list. Free Spaces run on shared GPU queues, so queue time is noted separately. Extra checks: mean brightness and Canny edge overlap against the resized original.

Scores are my own visual judgement; with only 3 images this is a qualitative comparison, not a benchmark.

4. Production Recommendation

Selected approach: own GPU running Qwen-Image-Edit-2509 with the Relight LoRA for time-of-day presets, behind an automatic quality gate and with a hosted preset tool as fallback; sun-direction control as a separate R&D track.

- **No tested option is ready for "move the sun".** Only Sun Direction LoRA moved the light, once (interior, 6° elevation), and it also switched lamps on, greyed the colors, added a chimney on the barn and returned smaller images. Every other run changed color, sky or brightness while the shadows stayed in place.
- **Time of day is solvable, and Relight Spaces sets the bar.** Its result on the sunset building (clear blue sky, all objects kept, full size, 39 s) was the best of all runs. It is a

real-estate photo web tool and I did not see an API on its site, so I would use it as a quality benchmark and a possible stopgap, not as a core dependency.

- **The open route has no license blocker.** Qwen-Image-Edit-2509 and its Relight LoRA are open-source with commercial use support, while the FLUX.2 klein 9B and FLUX.1 Kontext dev bases are non-commercial base models. Qwen kept geometry and lamps best on the barn and interior, but my runs returned smaller images (57–73% of the input width) and did not move the sun, so it needs trigger-word and prompt tests and a fixed output size before it can ship.

Comparison of the two deployment options:

	Hosted API	**Own GPU**
Cost per image	Relight Spaces $0.40–0.75 per photo; Magnific €0.10 per operation; fal IC-Light V2 $0.10 per megapixel (V2 not tested; IC-Light v1 failed on buildings).	Fixed compute infrastructure cost based on server provisioning and request load.
Speed	Relight Spaces 30–39 s (measured in the web UI).	Qwen LoRA 5–44 s on ComfyUI Cloud (GPU type unknown); Sun Direction 13–17 s on a Space. 15 s is typical.
Licensing	Provider handles it; fal lists IC-Light V2 for commercial use.	Qwen-Image-Edit-2509 + Relight LoRA: open-source with commercial use support. FLUX.2 klein 9B and FLUX.1 Kontext dev: non-commercial base models. IC-Light: swap its background remover.
Privacy of client projects	Images leave our infrastructure (Relight Spaces states uploads are private and not used for training, per its site).	Stays in our control.
Ops effort	Minimal.	Needs deployment, queueing, monitoring.

Appendix A. Practical Testing & Results

Full before/after images, prompts and settings for every test, per photo. Scores follow the rubric in section 3.

Image 1: Old Barn Exterior

Test goal: change the sun direction, keep thatched roof, walls and window unchanged.

Original image:



Tool 1: Sun Direction LoRA (FLUX.2 klein 9B), Hugging Face Space

Test	Sun position	Extra prompt	Seed	Time	Output size
1	top-left	bright daylight, clear sky	162	14 s	1248×832 (input 1920×1280)
2	bottom-left	none	162	13 s	1248×832 (input 1920×1280)



Test 1: sun top-left, extra prompt “bright daylight, clear sky”.



Test 2: sun bottom-left, no extra prompt.

What worked: the main geometry stayed in place in both tests (roof outline, chimney, wall, window, position of the tree). Shadows on the roof and ground look plausible and the lighting is cleaner and more even than in the original.

What failed (failure analysis):

- **Hallucinated structure.** In both tests a second white chimney and a roof-like shape appear behind the ridge on the right. In the original this area is only trees and sky (crop below). Same seed and same artifact in both tests suggests it comes from the model, not from the sun setting.



Area behind the roof ridge: original vs. Test 1 vs. Test 2.

- **Material and color drift.** The orange autumn trees in the background turn grey-brown, the dark apple-tree trunk becomes pale, and the ground leaves lose their warm color.
- **Weak light change in Test 1.** The original sun already comes low from the left, so “top-left” is close to the starting condition. Mean brightness rose from 85 to 138 (0–255), partly from the added “bright daylight” prompt, which also washed out the deep shadows of the original.

- **Test 2 did not look like light from below.** The result is flat and dim (mean brightness 108), with no visible upward lighting on the walls or roof. The model seems to map an unnatural direction to generic diffuse light.
- **Resolution loss.** The Space returns 1248×832 from a 1920×1280 input, which matters for architectural renders.

Edge check (Canny, 5 px tolerance, against the original resized to the same size): 73–74% of original edges are found in the result, and 65% of result edges match original edges. The mismatches include the added chimney and the changed foliage, so this is a rough indicator, not a clean geometry score.

Not done: Sun Direction tests with the sun from the right or from behind the building (clearly different from the original). The free Space ran out of GPU quota before they could be run.

Tool 2: Qwen-Image-Edit-2509 Relight LoRA (dx8152), ComfyUI Cloud

Setup: ComfyUI Cloud, workflow “Image Relight”

LoRA: dx8152/Qwen-Image-Edit-2509-Relight, strength 1.00, random seed. The template notes list the fp8 Qwen-Image-Edit-2509 base model and a Lightning 4-step LoRA next to the Relight LoRA; the Relight model card says it must be used together with a Lightning LoRA. The LoRA trigger-word node in my workflow was empty. The screen shows a run counter (“5 / 5 runs left”), so testing on the free cloud tier is limited.

Test	Light target	LoRA strength	Seed	Time	Output size
1	sun from the left	1.00	rand.	15 s	1248×832 (input 1920×1280)
2	sun from the right	1.00	rand.	5 s	1248×832 (input 1920×1280)
3	sun from the right, long prompt, blue sky	1.00	rand.	14 s	1248×832 (input 1920×1280)

Prompts used:

- **Test 1:** Preserve the exact architecture, geometry, windows, doors, materials, textures, furniture and camera viewpoint of the input image. Change only the lighting so that direct sunlight comes from the left side of the image, casting realistic shadows toward the left. Keep all objects and details unchanged.
- **Test 2:** Preserve the exact architecture, geometry, windows, doors, materials, textures, furniture and camera viewpoint of the input image. Change only the lighting so that direct sunlight comes from the right side of the image, casting realistic shadows toward the left. Keep all objects and details unchanged.
- **Test 3:** Relight the entire scene with direct sunlight coming from the RIGHT side of the image. The sun direction must visibly change compared with the original. Reverse the direction of the cast shadows so they extend toward the LEFT. Illuminate the right-facing surfaces of the building and leave the opposite-facing surfaces in shadow. Create clear, directional midday sunlight with a blue sky. Keep the building completely unchanged: preserve the exact roof shape, thatched roof, chimney, walls, windows, trees, ground, materials, colors and camera viewpoint. Do not add or remove objects. Change the lighting and shadows only.

Note on Test 1: the prompt asks for sun from the left but shadows toward the left, which is physically inconsistent (a sun on the left casts shadows to the right). The original sun already comes low from the left, so Test 1 is close to a no-change control.



Original vs. Test 1 (Qwen Relight LoRA): sun from the left, 15 s.



Original vs. Test 2 (Qwen Relight LoRA): sun from the right, short prompt, 5 s.



Original vs. Test 3 (Qwen Relight LoRA): sun from the right, long prompt with blue sky, 14 s.

What worked: geometry and objects stay in place in all three tests: roof outline, chimney, window, tree trunk and fence match the original (Canny edge recall 91–92%, precision 99%, against the original resized to 1248×832). No extra chimney or roof shape appeared, unlike the Sun Direction LoRA on the same image. The orange background trees and the leaves on the ground keep their color.

What failed (failure analysis):

- **No change of sun direction.** In all three tests the cast shadows (the long shadow across the ground, the shadow bands on the thatch) are in the same places as in the original; the 99% edge precision says almost no new edges appeared. Tests 2 and 3 asked for sun

from the right and reversed shadows, and the result still shows the original left-sun lighting. The longer, more explicit prompt in Test 3 did not help.

- **A color and contrast change instead.** Mean saturation (HSV, 0–255) goes 95 → 127 / 131 / 137 and mean brightness 85 → 77 / 77 / 71 for Tests 1 / 2 / 3; the roof gets darker (48 → 32 / 30 / 26). Test 3 gives a slightly deeper blue sky.
- **Material drift.** The thatch looks smoothed and more painterly than the original, and the background trees are blurrier.
- **Resolution loss.** 1920×1280 input, 1248×832 output (65% of the width), same as the Sun Direction Space on this image.
- **Likely cause (not tested yet).** The model card says the LoRA is driven either by prompts from its dataset or by the trigger word “重新照明”. My prompts were long edit instructions and the trigger-word node was empty, so the LoRA may act as a general color grade here instead of moving the sun.

Next tests for this tool: start the prompt with the trigger word “重新照明” and use the short style of the model card example (a clearly different light source), then repeat on Images 2 and 3, where a direction change is easier to see.

Tool 3: IC-Light (Foreground and Background Condition), Hugging Face Space

Space: Hugging Face IC-Light Space, Foreground and Background Condition mode. The light-source setting, seed and steps were not recorded.

Test	Light target	Light source setting	Seed	Time	Output size
1	a different sun direction (no side named)	not recorded	–	5 s	512×640 (input 1920×1280, cropped to 4:5)

- **Prompt:** Change the sun direction and resulting shadows to create clearly different, realistic sunlight. Preserve the exact original architecture, roof shape and thatch texture, walls, windows, materials, colors, and all objects. Do not add, remove, or redesign anything. Only change the sunlight direction and lighting. Keep the original camera angle and composition. The result must be the same photograph with different sunlight, not a regenerated scene.



Original (cropped to 4:5 as the Space does) vs. IC-Light Test 1, 5 s.

What worked: roof shape, chimney, thatch texture and the outlines of the tree and fence are kept, and nothing was added (edge precision 99%). It is the fastest tool so far (5 s).

What failed (failure analysis):

- **No change of sun direction.** Shadows on the thatch and on the ground are in the same places as in the original. The only lighting change is a darker, cooler image (mean brightness 91 → 64 on the cropped area).
- **Material and color drift.** The white plaster wall turns grey with a crack-like texture, the window gets smaller and loses its orange frame, the warm colors fade (average color 93/92/83 → 65/63/64) and the orange trees in the background look washed out.
- **Crop and resolution.** The Space center-crops the input to 4:5 and returns 512×640, so about 47% of the barn image width is lost and the rest is shown at half of the original pixel size. Aligned edge recall is only 65%, because detail disappears in the dark result.

Tool 4: Relight Spaces (commercial web tool, preset lighting styles)

Setup: relightspaces.com web app (“Scene Engine V2.5” according to the site). It offers fixed lighting styles (14 per the site) and no sun-direction control, so each photo was run once with the preset closest to the goal. 3 free relights without a card (the first one without an account), so no re-runs were possible. Price on the site: $0.40–0.75 per photo depending on the credit pack.

Test	Light target	Direction control	Seed	Time	Output size
1	preset “Golden Hour”	none (presets only)	n/a	36 s	1920×1280 (same as input)



Original vs. Relight Spaces, preset “Golden Hour”, 36 s.

What worked: geometry and objects stay in place (Canny edge recall 96%, precision 80%) and nothing was added or removed. Thatch, plaster, window frame and fence keep their texture, and the output has the size of the input. The scene turns into a golden evening: the orange foliage glows, the sky becomes pink-gold (mean saturation 99 → 150).

What failed (failure analysis):

- **No change of sun direction.** The long ground shadow, the tree shadows and the shadow bands on the thatch are in the same places as in the original. The preset changes color and sky, not where the light comes from. Mean brightness is unchanged (85 → 84).
- **Mild test by my choice.** The original already has low warm sun, so “Golden Hour” is a small change here. A preset closer to a time-of-day change was not tried, because only 3 free relights were available.

Scores (1–5, rubric in section 3):

Tool	Geom.	Mat.	Light	Shad.	Time	Notes / failure analysis
Sun Direction, Test 1	4	3	3	4	14 s	Extra chimney/roof added; colors drift; direction change small
Sun Direction, Test 2	4	3	2	3	13 s	Same added structure; flat light, no light from below
Qwen Relight LoRA, Tests 1–3	5	3	1	1	5–15 s	Geometry kept, nothing added; but sun direction unchanged, colors more saturated, thatch smoothed
IC-Light, Test 1	3	2	1	1	5 s	Roof and thatch kept, but wall turns grey, image darker, shadows unchanged; 4:5 crop at 512×640
Relight Spaces, Golden Hour	5	4	3	2	36 s	Everything kept at full size; warm golden grade and pink sky, but shadows in the same places

Image 2: Modern Building / Sunset Exterior

Test goal: change from sunset to midday daylight, keep wall cladding, roof and windows unchanged.

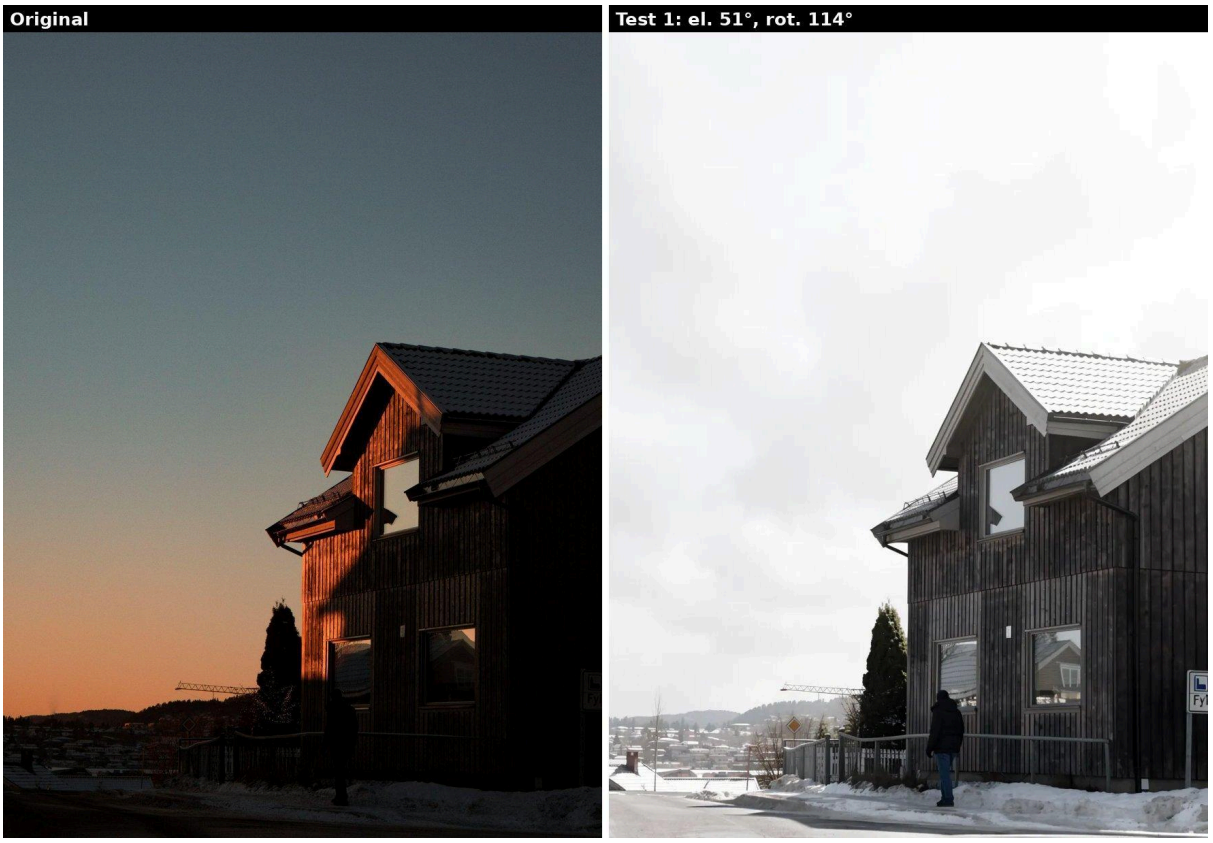
Original image:



Prompt / light target used: Sun Direction LoRA (FLUX.2 klein 9B) via the Hugging Face Space, sun-position control only. Test 1: elevation 51°, rotation 114° (sun top-right). Test 2: sun top-right, elevation and rotation not noted. Extra prompt: none.

Test	Sun position	Extra prompt	Seed	Time	Output size
1	top-right (el. 51°, rot. 114°)	none	162	17 s	880×1184 (input 1536×2048)
2	top-right (el./rot. not noted)	none	162	16 s	880×1184 (input 1536×2048)

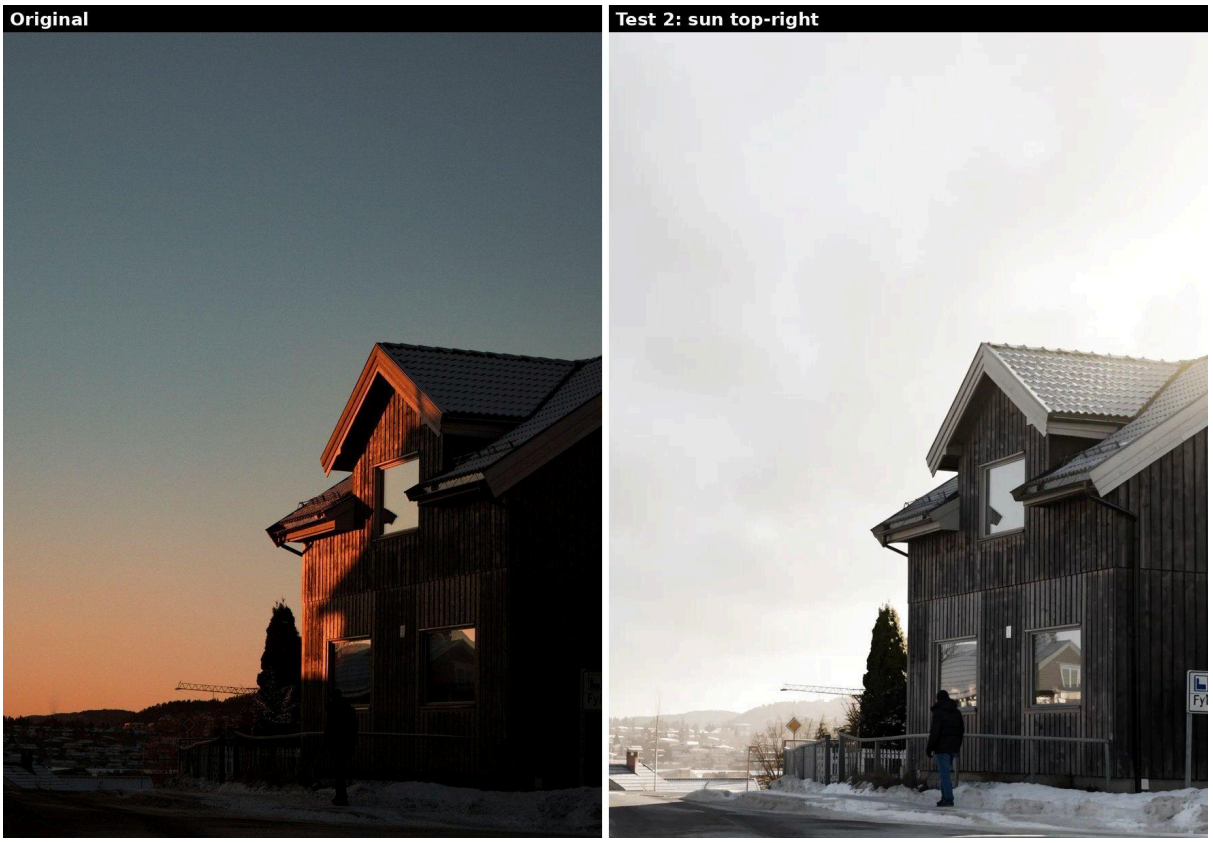
Successful attempt



Original vs. Test 1: sun top-right, elevation 51°, rotation 114°, seed 162, 17 s.

- **What worked:** the sunset look is gone. The orange wall light and the orange-teal sky are replaced by flat neutral light that reads as midday (mean brightness 78 → 188 on a 0–255 scale). Gable, roof lines, the three windows, the vertical cladding, the tree and the fence stay in place (Canny edge recall 88%).
- **What is still wrong:** the sky is blank overcast white, not the blue sky a midday goal implies. The light is diffuse, with no clear shadow from a top-right sun, so it works as a time-of-day change but not as a sun-direction change. The small lights on the tree disappear, the dark roof tiles turn pale and frosty, and the dark silhouette of the person by the fence is redrawn as a detailed figure in a jacket. Edge precision is only 31% because snow and roof texture add many new edges, so the edge check is a rough indicator only.
- **Resolution loss:** the Space returns 880×1184 from a 1536×2048 input.

Failed attempt (failure analysis)



Original vs. Test 2: sun top-right (elevation and rotation not noted), seed 162, 16 s.

- **What failed:** the time-of-day change is incomplete. The result keeps a golden-hour feel: a warm hazy glow at the top-right and warm haze across the sky, with no blue sky and no midday light (mean brightness 177). Geometry is as good as in Test 1 (edge recall 88%), and the same side effects appear: tree lights removed, pale roof, redrawn person.
- **Open question:** elevation and rotation were not noted for this run, so I cannot say if a lower elevation caused the warm look. The next run should use the same elevation as Test 1 with an extra prompt such as “midday sun, bright blue sky”.

Tool 2: Qwen-Image-Edit-2509 Relight LoRA (dx8152), ComfyUI Cloud

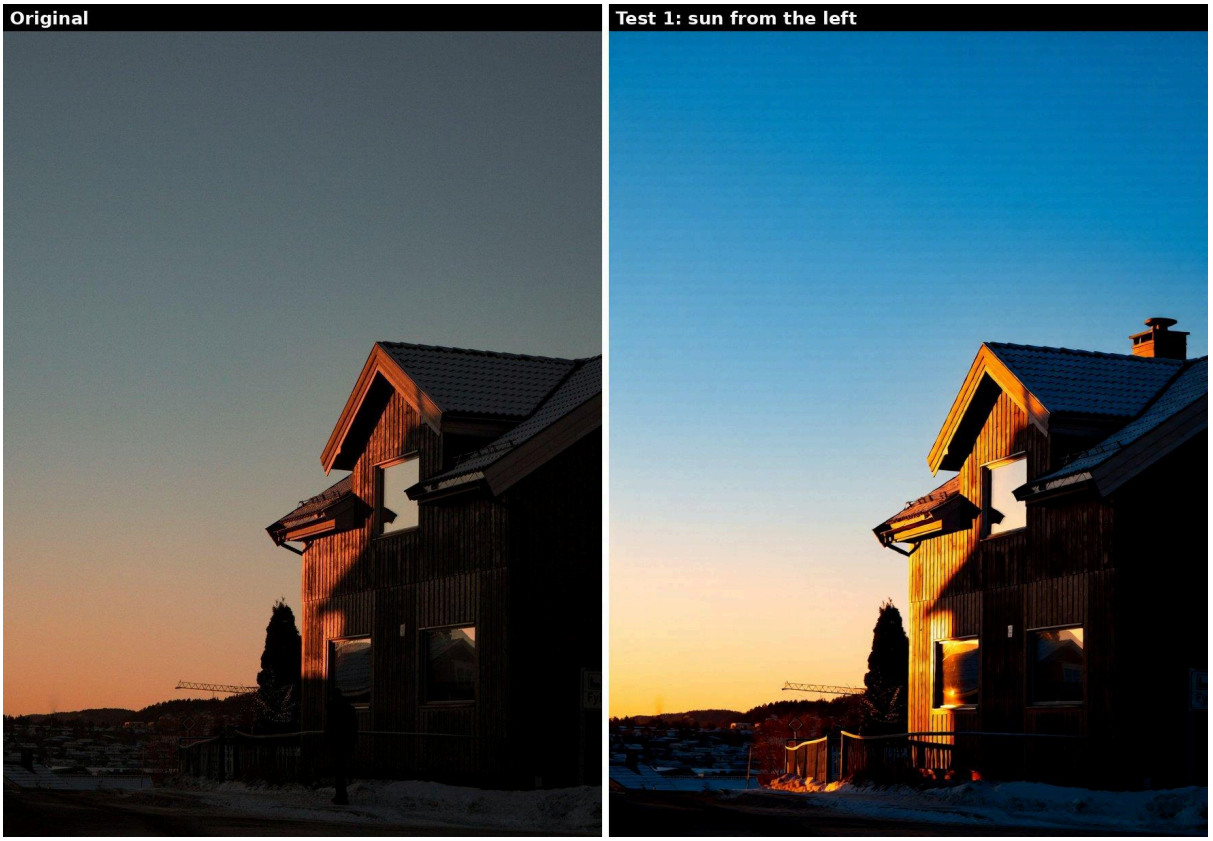
Setup: same ComfyUI Cloud workflow and LoRA as for Image 1 (see above), three prompts of increasing strength.

Test	Light target	LoRA strength	Seed	Time	Output size
1	sun from the left (original direction)	as Image 1	rand.	14 s	880×1184 (input 1536×2048)
2	sun from the right, midday, blue sky	as Image 1	rand.	44 s	880×1184 (input 1536×2048)
3	sun from the right, midday, strict preservation	as Image 1	rand.	18 s	not kept

Prompts used:

- **Test 1:** Relight the entire scene with direct sunlight coming from the LEFT side of the image. The sun direction must visibly change compared with the original. Reverse the direction of the cast shadows so they extend toward the RIGHT. Keep the building completely unchanged: preserve the exact roof shape, thatched roof, chimney, walls, windows, trees, ground, materials, colors and camera viewpoint. Do not add or remove objects. Change the lighting and shadows only.
- **Test 2:** Change the sun direction from LEFT to RIGHT. This is the primary requirement. Replace the existing warm sunset lighting with strong, direct midday sunlight coming clearly from the RIGHT side of the image. The right-facing surfaces of the building must receive bright direct sunlight. The left-facing surfaces must be in shadow. Reverse the existing cast-shadow direction so shadows extend toward the LEFT. Change the orange sunset sky to a clear natural blue midday sky. Remove the golden-hour glow and warm atmospheric haze. Create distinct, physically consistent directional shadows. Preserve the exact building geometry, roof, windows, vertical facade cladding, fence, trees, person and all other objects. Do not add, remove, move or redesign anything. Keep the original camera viewpoint and composition. The final image must clearly show sunlight coming from the opposite side compared with the input. Do not merely change brightness, color temperature or sky color.
- **Test 3 (shortened here):** Relight the existing image. This is a lighting-editing task, NOT a scene-generation task. Change only the sunlight direction and lighting conditions. Move the apparent sun direction from the LEFT to the RIGHT side of the image. [...] Replace the warm sunset lighting with neutral, natural midday sunlight and a clear blue sky. Strict scene preservation: preserve the exact original building and all its architectural details; do not add any buildings, structures, chimneys, roofs, windows, doors or other objects; do not remove, move, redesign or duplicate anything; keep the original camera angle, perspective, framing and composition. [...] Priority order: 1. Preserve scene geometry. 2. Reverse sunlight direction. 3. Change sunset to midday.

Note: the original sun already comes from the left, so Test 1 is a weak request (same direction, reversed shadows). Tests 2 and 3 ask for the opposite direction and a midday look.



Original vs. Test 1 (Qwen Relight LoRA): sun from the left, 14 s.



Original vs. Test 2 (Qwen Relight LoRA): sun from the right, midday, blue sky, 44 s.

What worked: Test 1 keeps most of the scene in place (Canny edge recall 97%): gable, roof, windows, cladding, crane, tree and fence. The sky becomes a clean blue-to-gold gradient and the left-facing wall, eaves and windows catch stronger orange sun. Test 2 does remove the golden-hour glow: deep blue sky, no warm haze, and the main facade is lit by direct sun with a diagonal shadow line across it. The vertical cladding, windows and roof of the main building stay where they were.

What failed (failure analysis):

- **Test 1: the sunset stays.** It is the original light with more saturation (mean saturation 84 → 171), so no midday and no new direction. A brick-like chimney appears on the roof that is not in the original, and the person by the fence and the lights on the tree are gone (edge precision 71%).
- **Test 2: the scene is regenerated.** A whole second house with windows and a snowy roof appears on the left, replacing the distant town, hills and crane; the tree and the person are gone too. The prompt said “do not add, remove, move or redesign anything”. Edge recall falls to 85% and precision to 42%.
- **Test 2: the direction is not as requested.** The wall that faces left is now fully lit, while the prompt asked for the left-facing surfaces to be in shadow. The sky is extremely saturated and looks artificial (mean saturation 195). It took 44 s, about three times Test 1.
- **Test 3: almost no change.** With the strictest “preserve everything” prompt (18 s) the result barely differed from the original, so I did not keep the image; no scores are given for it.
- **Takeaway.** Prompt strength decides the outcome: a mild prompt keeps the golden hour, a strong midday prompt regenerates the scene, and a strict preservation prompt changes nothing. I found no prompt in between that changed the light and kept the scene.

Resolution: 880×1184 output from a 1536×2048 input, the same loss as with the Sun Direction Space.

Tool 3: IC-Light (Foreground and Background Condition), Hugging Face Space

Space: Hugging Face IC-Light Space, Foreground and Background Condition mode. The light-source setting, seed and steps were not recorded.

Test	Light target	Light source setting	Seed	Time	Output size
1	sun from the right, neutral midday, blue sky	not recorded	–	7 s	512×640 (input 1536×2048, cropped to 4:5)

- **Prompt:** Change the sunlight direction from LEFT to RIGHT. Replace warm sunset lighting with strong, neutral midday sunlight coming from the RIGHT, casting shadows toward the LEFT. Change the orange sky to a natural blue sky. Preserve the exact building geometry, facade, windows, materials, colors, trees, fence and all objects. Do not add, remove or redesign anything. Keep the original camera angle and composition. Only change sunlight direction, lighting and shadows. The result must be the same photograph with different lighting.



Original (cropped to 4:5 as the Space does) vs. IC-Light Test 1, 7 s.

What worked: the building is kept almost untouched: roof, windows, cladding, crane, tree, fence and the person by the fence are in place (edge recall 98%). Nothing was added or removed, and it took 7 s.

What failed (failure analysis):

- **The building light does not change.** The left face is still lit orange by the low sun and the shadow lines are the same as in the original, so neither the direction nor the time of day changed on the building itself.
- **Only the background was repainted.** The orange-teal sky was replaced by a smooth gradient, light on the left and almost black on the right, with visible vertical banding. There

is no blue sky, the sunset glow stays at the horizon, and the dark gradient does not match the lit foreground, so the result looks like a cut-out on a new background.

- **Crop and resolution.** 512×640 from a 1536×2048 input (cropped to 4:5), about a third of the original size.

Tool 4: Relight Spaces (commercial web tool, preset lighting styles)

Test	Light target	Direction control	Seed	Time	Output size
1	preset “Wide Awake” (sunset → day)	none (presets only)	n/a	39 s	1920×2560 (input 1536×2048)



Original vs. Relight Spaces, preset “Wide Awake”, 39 s.

What worked: the sunset becomes a clear blue-sky day (mean brightness 78 → 144; average color grey 79/78/74 → blue-leaning 119/149/181). Gable, roof, windows, vertical cladding, crane, tree with its lights, fence, road sign and the person are all in place (Canny edge recall 91%). The shadow on the facade stays where it was, so lit and shaded sides remain consistent. The output is 1920×2560, 1.25× larger than the input, the opposite of the resolution loss on the Spaces.

What failed (failure analysis):

- **No sun-direction control.** Only a preset: the light follows the original shadow layout, so this is a time-of-day change, not a direction change.
- **Detail reconstructed in dark areas.** The person was a dark silhouette and now shows a jacket and trousers; snow, cladding and window reflections (neighbour roofs) appear where the original was almost black. It looks plausible, but it is detail the original did not contain. Edge precision is only 17% because these areas now have many new edges, so the edge check is a rough indicator only.

Scores (1–5, rubric in section 3):

Tool	Geom.	Mat.	Light	Shad.	Time	Notes / failure analysis
Sun Direction, Test 1	4	3	4	2	17 s	Midday-like neutral light, orange gone; sky white not blue; no clear directional shadows; tree lights lost
Sun Direction, Test 2	4	3	2	3	16 s	Still golden-hour haze; time-of-day change incomplete; same material drift
Qwen Relight, Test 1	3	3	2	3	14 s	Golden hour stays, more saturation; chimney added, person removed
Qwen Relight, Test 2	1	2	3	3	44 s	Midday sky and lit facade, but second house invented, crane/tree/person removed, direction not as requested
Qwen Relight, Test 3	–	–	–	–	18 s	Almost no change; image not kept
IC-Light, Test 1	5	4	1	1	7 s	Foreground untouched, sky replaced by a gradient; sun direction and sunset light on the building unchanged
Relight Spaces, Wide Awake	5	4	4	3	39 s	Sunset → clear blue day, all objects kept, 1920×2560; no direction control; dark areas get reconstructed detail

Image 3: Interior Room with Window Blinds

Test goal: move the light direction, keep blind slats, furniture and layout unchanged.

Original image:



Prompt / light target used: Sun Direction LoRA (FLUX.2 klein 9B) via the Hugging Face Space, sun-position control only. Test 1: elevation 36°, rotation −90° (sun top-left). Test 2: elevation 6°, rotation −112° (sun far left, very low). In the original the sunlit patches fall on the left wall and left floor. Extra prompt: none.

Test	Sun position	Extra prompt	Seed	Time	Output size
1	top-left (el. 36°, rot. −90°)	none	162	14 s	1360×768 (input 1920×1080)
2	far left, low (el. 6°, rot. −112°)	none	162	14 s	1360×768 (input 1920×1080)

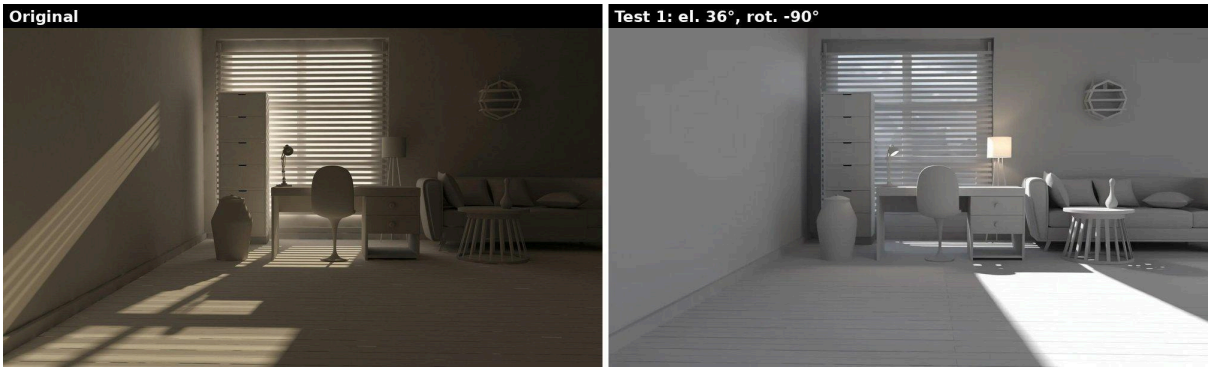
Successful attempt



Original vs. Test 2: sun far left and low, elevation 6°, rotation −112°, seed 162, 14 s.

- **What worked:** the light direction changed as requested. The patches moved from the left wall and left floor to the right side of the floor, and they keep the thin stripe pattern of the blind slats. Blinds, shelf unit, desk, chair, sofa and wall decor stay in place. Mean brightness 73 → 115.
- **What is still wrong:** the warm sepia light became neutral grey (average color about 81/72/60 → 115/115/115). The standing lamp, which is off in the original, is now lit. The light patch is cut by the right edge of the frame, and the left wall lost its slat shadows. Edge recall is 54%, but this is partly expected: the original's own light-patch edges moved, so the number is not a clean geometry score.
- **Resolution loss:** the Space returns 1360×768 from a 1920×1080 input.

Failed attempt (failure analysis)



Original vs. Test 1: sun top-left, elevation 36°, rotation −90°, seed 162, 14 s.

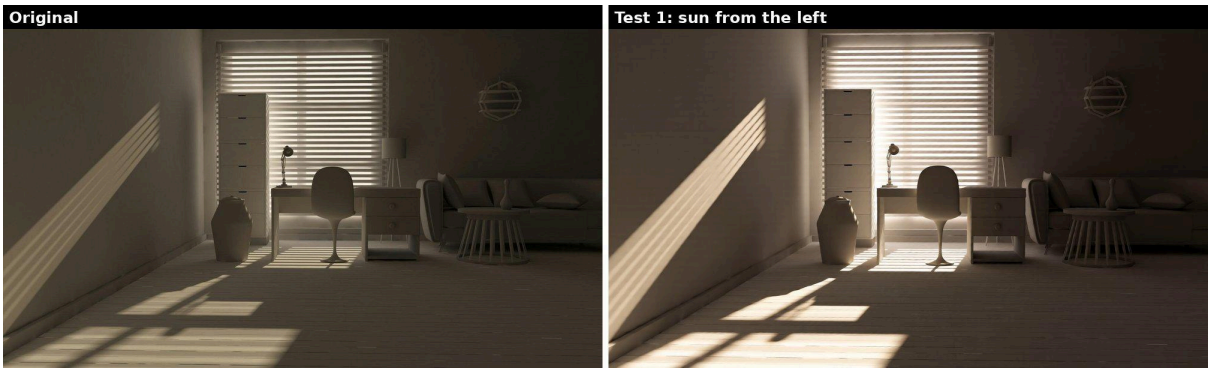
- **What failed:** the sun patch on the floor is one large soft-edged shape with no blind-slat stripes, so the fine structure of the blinds is lost and the shadow no longer matches the window. The floor lamp is lit again, the colors turn neutral grey (mean brightness 125) and the slat shadows on the left wall are gone. Edge recall 55%.
- **Takeaway:** a high sun elevation (36°) smears the light; a low elevation (6°) kept the stripes. An elevation between the two is worth a test.

Tool 2: Qwen-Image-Edit-2509 Relight LoRA (dx8152), ComfyUI Cloud

Setup: same ComfyUI Cloud workflow and LoRA as for Image 1 (see above), one test.

Test	Light target	LoRA strength	Seed	Time	Output size
1	sun from the left, beams through the blinds	as Image 1	rand.	15 s	1392×752 (input 1920×1080)

- **Prompt:** Relight the existing interior photograph. Change only the direction of sunlight. Direct sunlight enters through the existing window from the LEFT side of the image and travels diagonally toward the RIGHT side of the room. Create clearly visible, realistic parallel sunbeams passing through the existing horizontal window blinds. The beams should cast distinct striped patterns of light and shadow across the existing interior surfaces. Keep the original neutral interior colors and realistic daylight balance. Make the direction of the sunlight clearly visible through the light beams and shadow patterns. Preserve the exact room geometry, window, horizontal blinds, walls, ceiling, furniture, lamps and all existing objects. Do not add, remove, move, duplicate or redesign anything. Do not change the camera viewpoint or composition. Do not turn on any lamps. This is a controlled lighting edit, not a scene redesign. Only modify sunlight direction, illumination and the resulting shadows.



Original vs. Test 1 (Qwen Relight LoRA): sun from the left, beams through the blinds, 15 s.

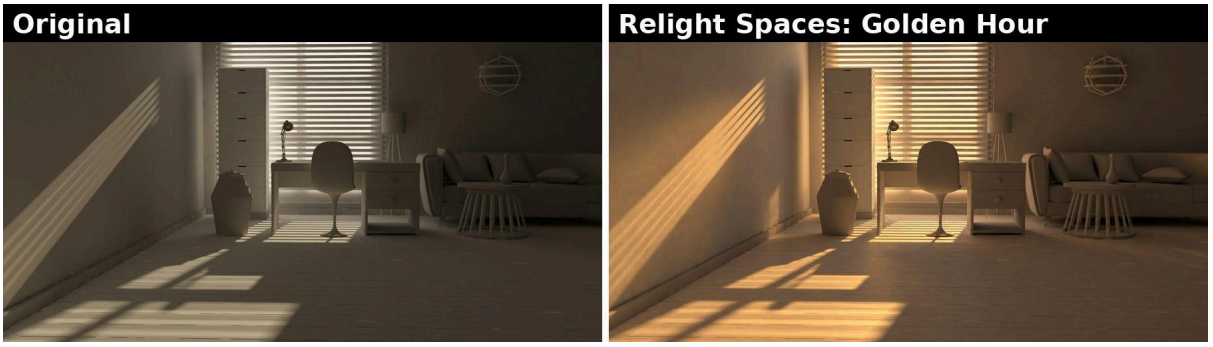
What worked: the scene is fully preserved: furniture, blinds, shelf unit, wall decor and layout are in the same places (Canny edge recall 90%). The lamps stay off, unlike the Sun Direction LoRA, and the warm interior colors are kept without a grey shift (average color 81/72/60 → 98/86/74). The light patches and blind stripes become brighter and sharper (mean brightness 73 → 89; share of very bright pixels on the floor 0.7% → 18%).

What failed (failure analysis):

- **No change of light direction.** The diagonal stripe on the left wall and the patches on the left part of the floor are in the same places as in the original. The result is the original light with more exposure, although the prompt asked for beams entering from the left and travelling to the right.
- **Aspect ratio and resolution.** 1920×1080 input, 1392×752 output (73% of the width, ratio 1.78 → 1.85, probably size rounding in the workflow), so the image is slightly stretched.
- **Takeaway.** On this interior the Qwen LoRA is safe (no added objects, lamps off, colors kept) but it does not move the sun, as on the barn. The Sun Direction LoRA at a low elevation did move the light, at the cost of colors and lamps.

Tool 4: Relight Spaces (commercial web tool, preset lighting styles)

Test	Light target	Direction control	Seed	Time	Output size
1	preset “Golden Hour”	none (presets only)	n/a	30 s	1920×1080 (same as input)



Original vs. Relight Spaces, preset “Golden Hour”, 30 s.

What worked: the scene is fully preserved (edge recall 97%, precision 76%): furniture, blinds and wall decor are in place, the lamps stay off and the output keeps the input size. The grey-sepia light becomes warm gold (average color 81/72/60 → 116/90/64, mean brightness 73 → 95) and the blind stripes stay sharp.

What failed (failure analysis):

- **No change of light direction.** The stripe on the left wall and the patches on the floor are in the same places as in the original. It is the original light with a stronger warm tone, similar to the Qwen result on this image.
- **Preset choice.** The site lists “Golden Hour” as best for exteriors and “Golden Air” as best for interiors. I used Golden Hour here, and Golden Air was not tried (free relights used up).

Scores (1–5, rubric in section 3):

Tool	Geom.	Mat.	Light	Shad.	Time	Notes / failure analysis
Sun Direction, Test 1	4	3	3	2	14 s	Direction moved, but slat pattern lost in a soft floor patch; lamp switched on; colors grey
Sun Direction, Test 2	4	3	4	4	14 s	Light moved to the right with slat stripes kept; lamp on, colors grey, patch cut by frame
Qwen Relight, Test 1	5	4	2	2	15 s	Scene and colors kept, lamps off; light patches not moved, only brighter
IC-Light	–	–	–	–	–	Not tested on this image
Relight Spaces, Golden Hour	5	4	2	2	30 s	Scene, lamps and size kept, warmer tone; light patches not moved